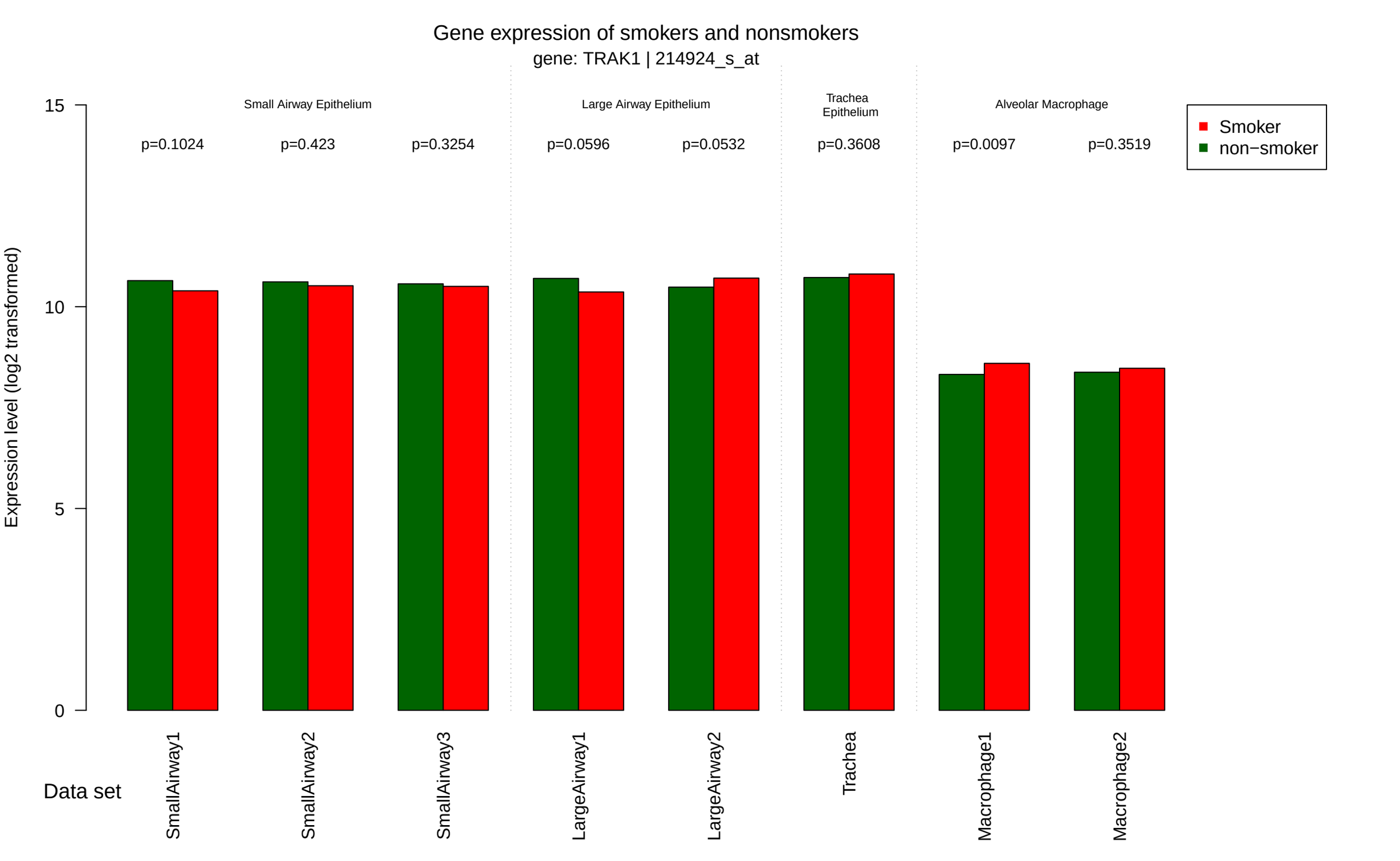
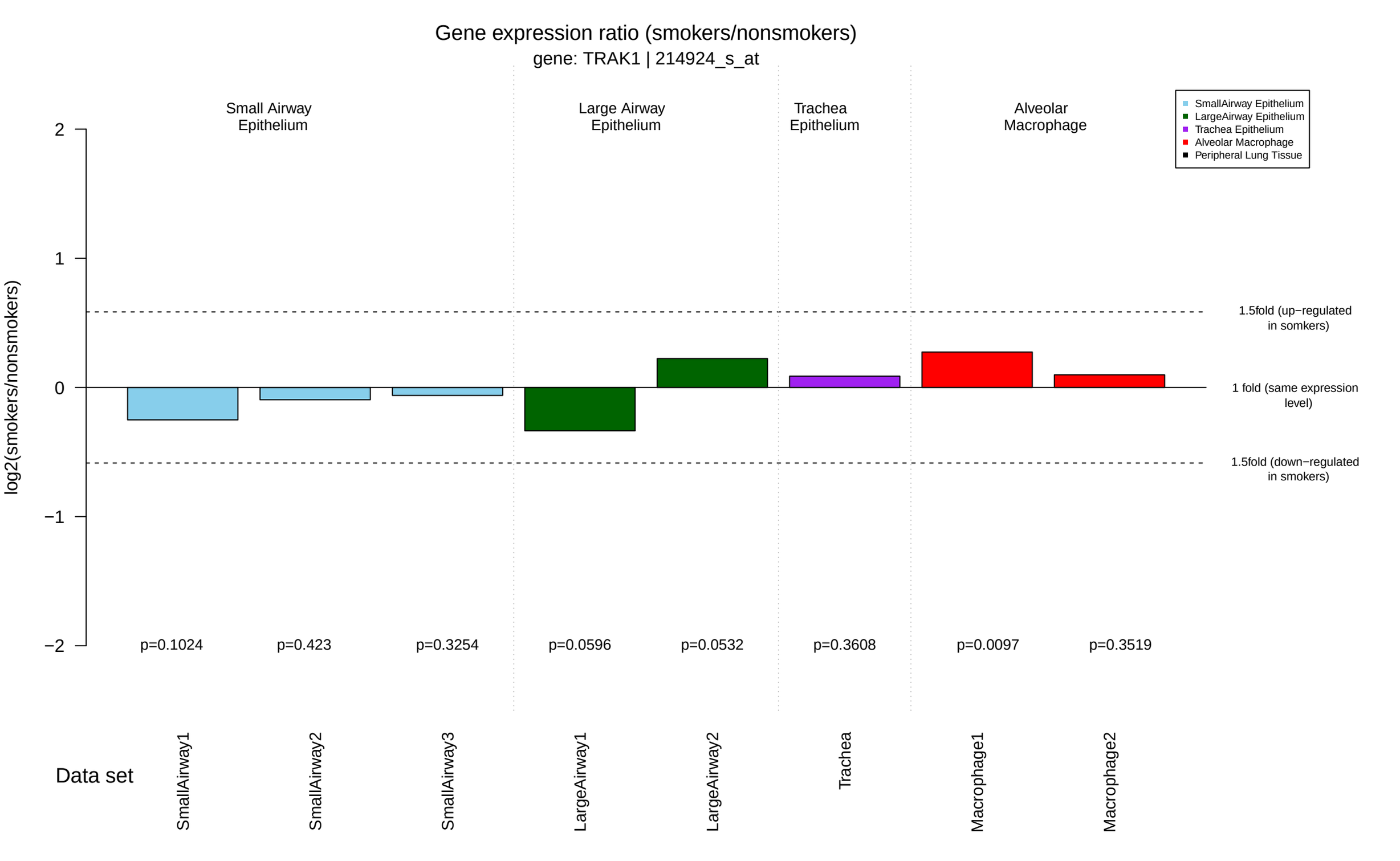
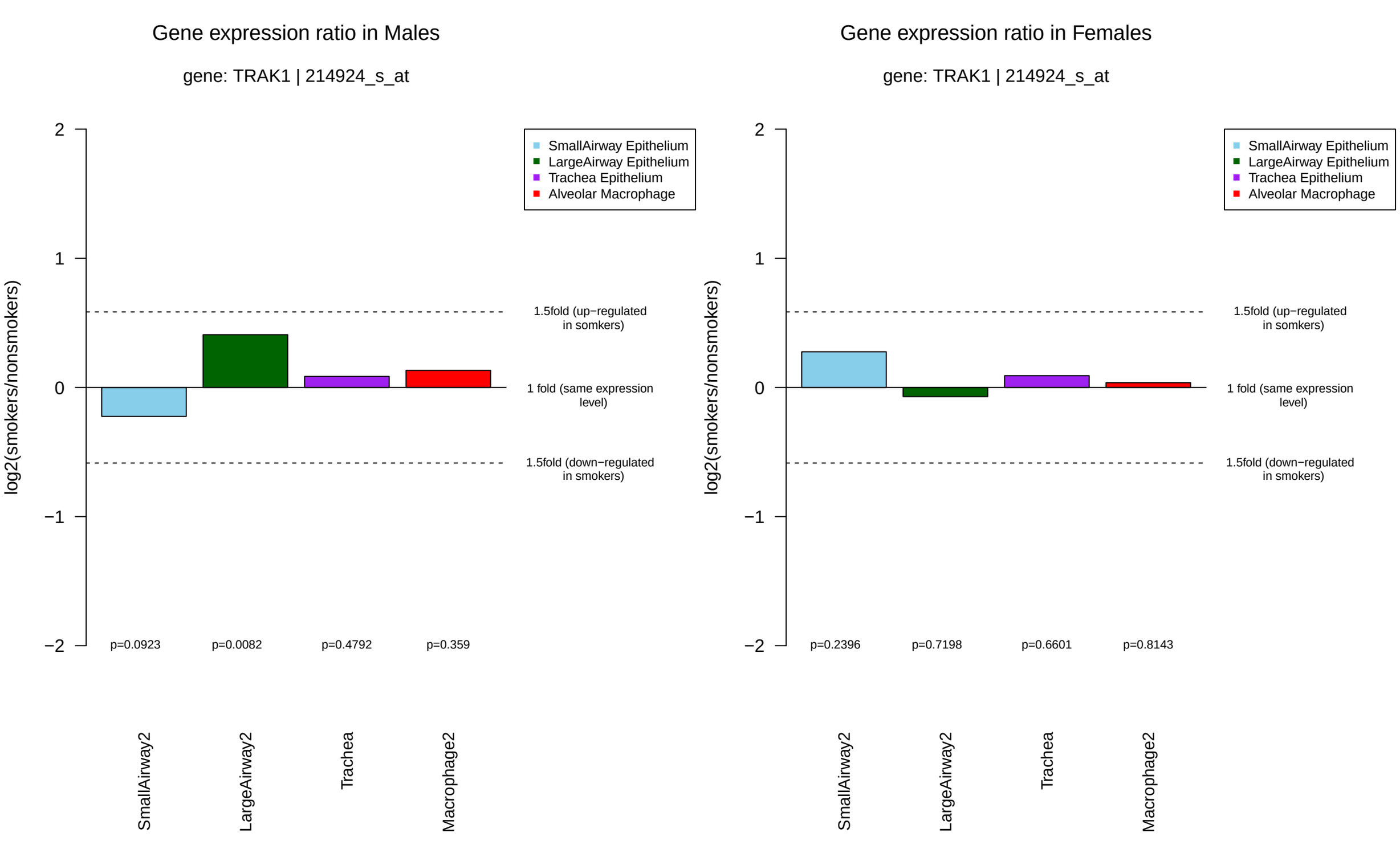
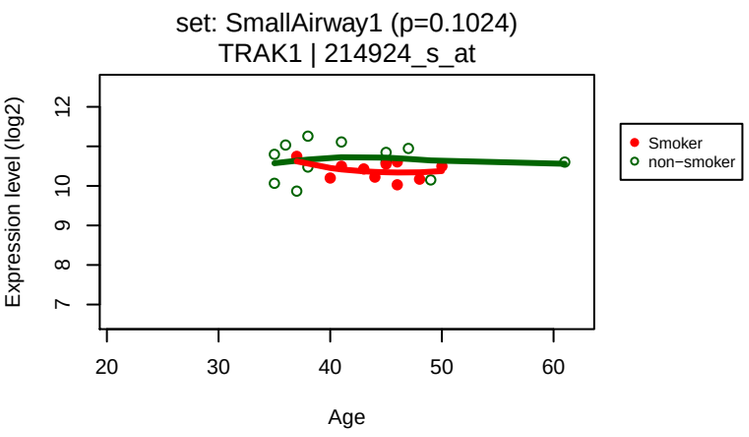
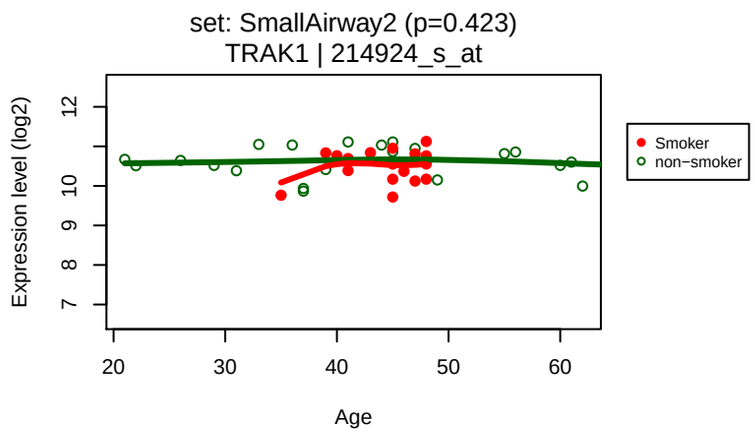
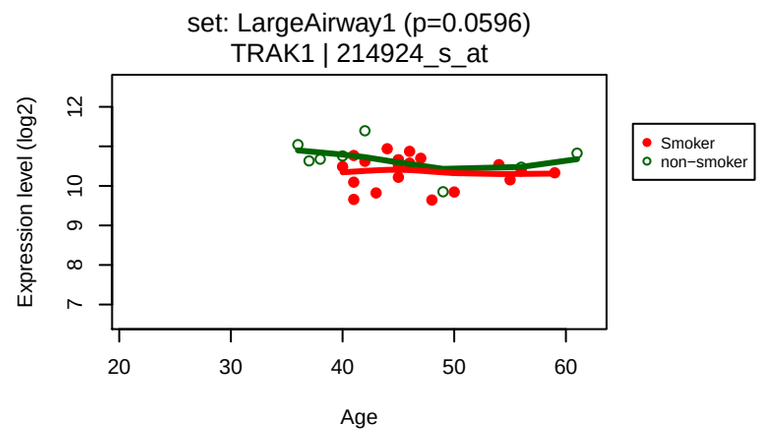

Gene expression of smokers and nonsmokers
gene: TRAK1 | 214924_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1024
p=0.423
p=0.3254
p=0.0596
p=0.0532
p=0.3608
p=0.0097
p=0.3519
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

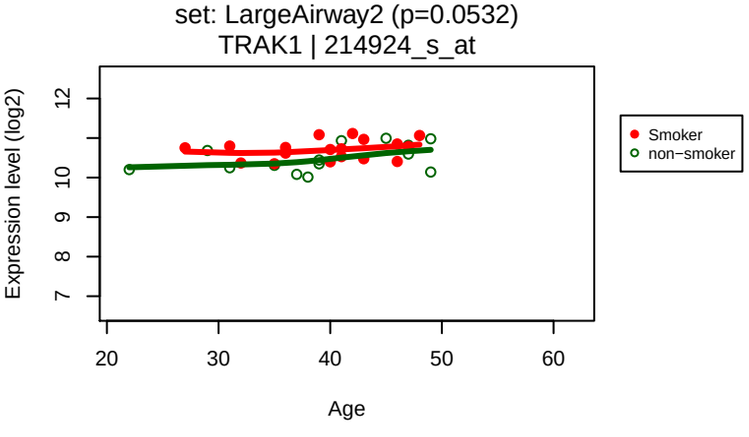

Gene expression ratio (smokers/nonsmokers)
gene: TRAK1 | 214924_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1024
p=0.423
p=0.3254
p=0.0596
p=0.0532
p=0.3608
p=0.0097
p=0.3519
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

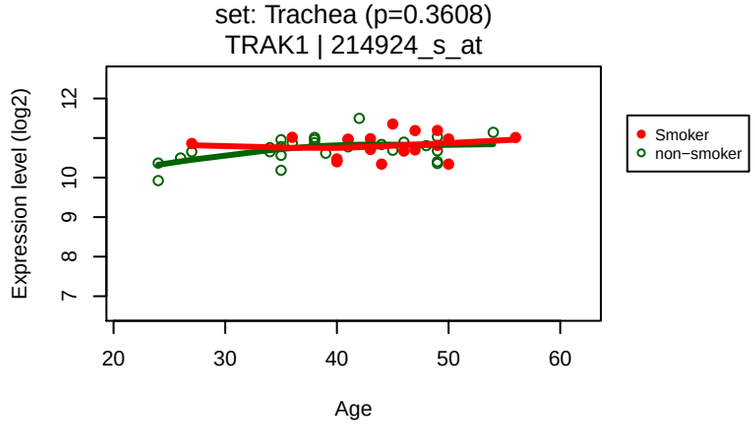

Gene expression ratio in Males
gene: TRAK1 | 214924_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0923
p=0.0082
p=0.4792
p=0.359
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRAK1 | 214924_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2396
p=0.7198
p=0.6601
p=0.8143
SmallAirway2
LargeAirway2
Trachea
Macrophage2

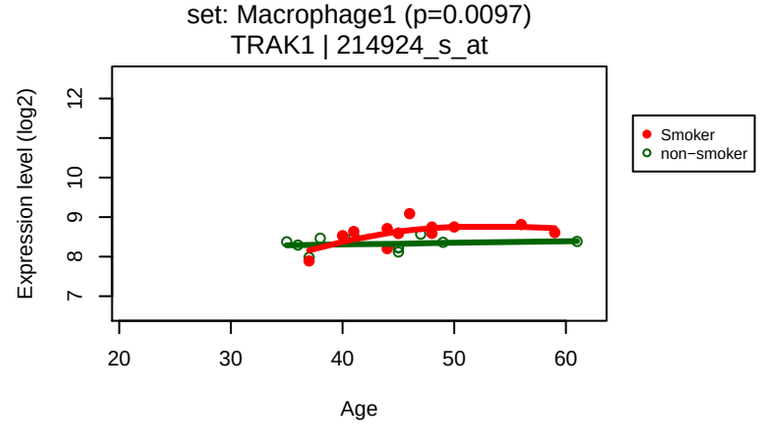

set: SmallAirway1 (p=0.1024)
TRAK1 | 214924_s_at
Expression level (log2)
Age
Smoker
non−smoker

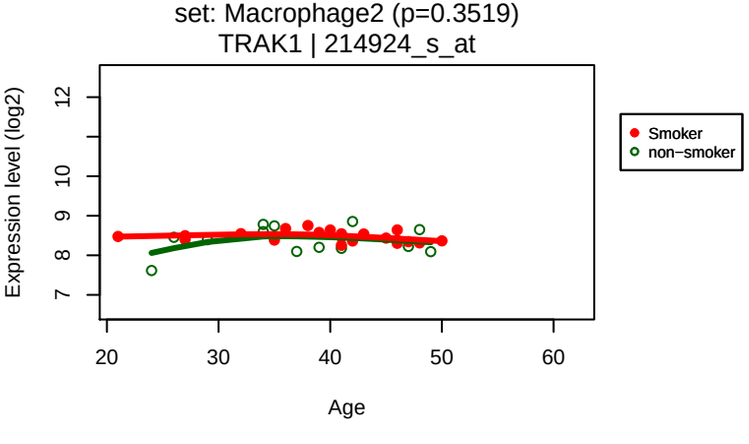

set: SmallAirway2 (p=0.423)
TRAK1 | 214924_s_at
Expression level (log2)
Age
Smoker
non−smoker

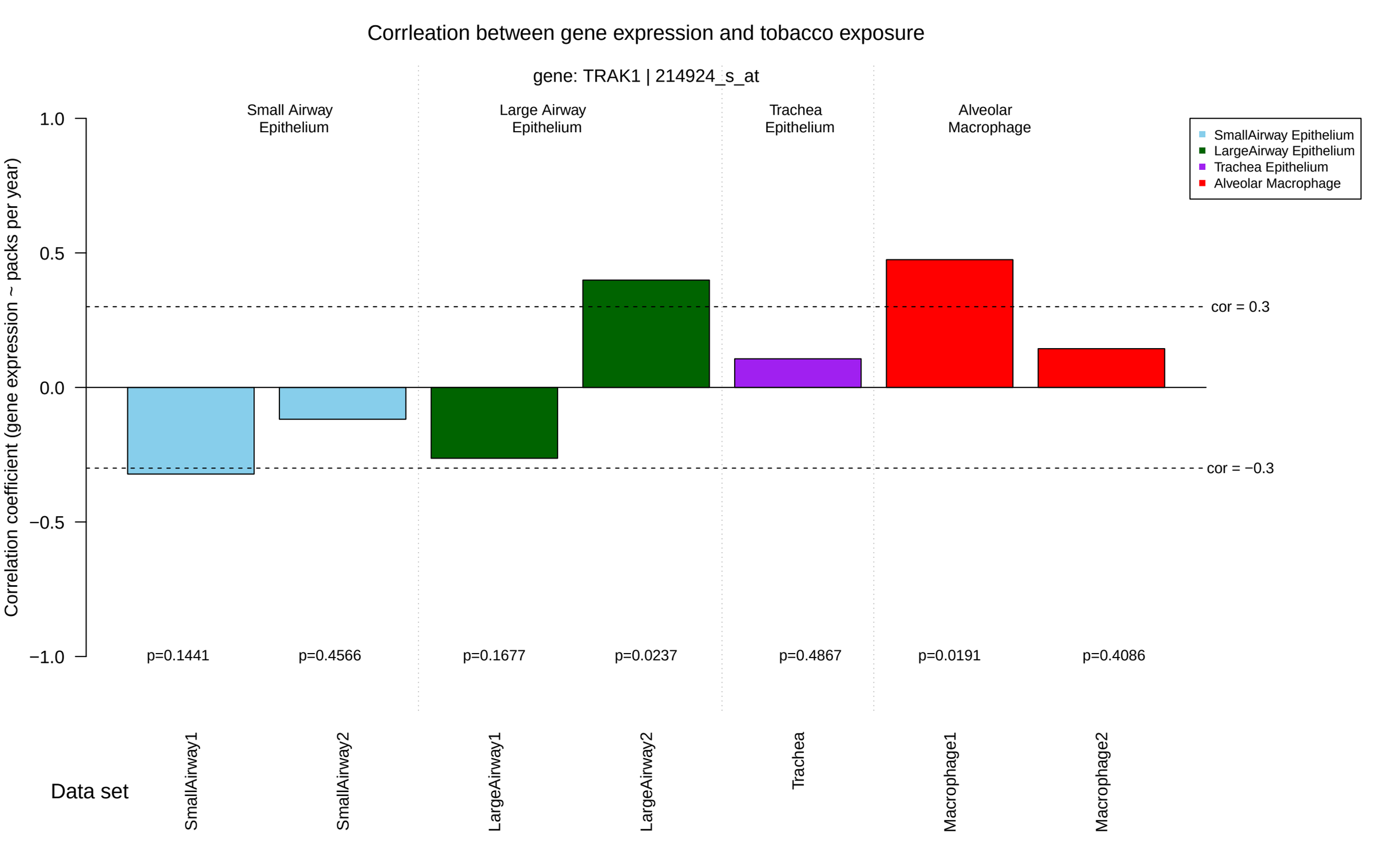

set: LargeAirway1 (p=0.0596)
TRAK1 | 214924_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0532)
TRAK1 | 214924_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3608)
TRAK1 | 214924_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0097)
TRAK1 | 214924_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3519)
TRAK1 | 214924_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRAK1 | 214924_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1441
p=0.4566
p=0.1677
p=0.0237
p=0.4867
p=0.0191
p=0.4086
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2